

Abdur Rahman Moayed

Software Engineer · Browser Internals, Systems Security & Web Architecture

Dhaka, Bangladesh (Remote Available) Email: armabdur.rahman04@gmail.com

GitHub: github.com/watchknight LinkedIn: [in/abdur-rahman-moayed-9225b5389](https://in.linkedin.com/in/abdur-rahman-moayed-9225b5389)

EDUCATION & ACADEMIC DISTINCTIONS

East West University (EWU) — *B.Sc. in Computer Science & Engineering* 2025 – 2029 (Expected)

- **Undergraduate Scholar:** Specializing in Systems Architecture, Operating Systems, Browser Internals, and Distributed Web Engineering.
- **Academic Chronology:** Completed Higher Secondary Certificate (HSC Science, 2023).
- **Unbroken Board Distinction:** Maintained consistent GPA 5.0 / 5.0 across Class 10 (SSC, 2021), Class 8 (JSC, 2018), and Class 5 (PSC, 2015) national board examinations.
- **Board General Scholarships:** Awarded prestigious Board General Merit Scholarships in Class 8 (JSC) and Class 10 (SSC) for regional academic excellence.

TECHNICAL PROJECTS

PureFeed — Browser Engine & Anti-Adblock Countermeasures JavaScript, Chromium Manifest V3, HTML5 Media

- Engineered zero-latency hybrid blocker bypassing YouTube's recursive 20ms playback loop with single-entry state guard and in-player 16x acceleration.
- Overcame strict host CSP (`script-src 'self'`) via Manifest V3 Main World registration, using the HTML root dataset as a zero-latency cross-world bridge.
- Eliminated Facebook's sponsored post evasion using recursive tree-walker with Unicode zero-width stripping, anagram frequency matcher, and 300-entry FIFO cache maintaining 60fps.

FocusGuard — Multi-Layer Network & OS-Level Policy Enforcement Chrome Extensions, Windows Registry, System APIs

- Architected 5-tier tamper-resistant content filter enforcing system-wide rules across Chrome, Edge, Firefox, and desktop applications.
- Synchronized browser declarativeNetRequest rules with OS hosts file (580+ domains redirected to `0.0.0.0`) and enterprise registry policy locks.
- Built an automated background watchdog service executing every 60 seconds to detect modifications and restore policy rules.

Rannabanna — Heuristic Matchmaking Engine & Culinary Physics ([Repo](#) · [Live Demo](#)) React, Node.js, SQLite, Gemini 3.8 Flash API

- Designed weighted set-intersection matchmaking engine across 395+ recipes and 182 standardized ingredients, eliminating combinatorial explosion.
- Implemented non-linear physics scaling formulas: prep duration scaling by square root of serving ratio ($\text{prep} \times \sqrt{r}$) and cook time logarithmic damping.
- Built tiered translation pipeline with Gemini 3.8 Flash, SHA-256 persistent database caching, and sub-5ms memory lookup to avoid duplicate API calls.

System ([Repo](#) · [Live Demo](#))

React, TypeScript, Tailwind CSS, MediaPipe FaceMesh

- Developed high-performance eyewear storefront with instantaneous client-side faceted filtering by frame shape, material, and lens dimensions with zero layout shift.
- Integrated MediaPipe 468-point 3D Face Mesh virtual try-on with real-time landmark interpolation at 60 FPS.

Live Demo)

Next.js 16, React 19, Supabase, Zustand,
SSLCommerz

- Engineered end-to-end commerce interface featuring atomic cart state synchronization, resilient local persistence, and idempotent regional payment verification.

(Repo · Live Site)

React 19, TypeScript, Vite, Web Audio API, Canvas 2D, Lenis

- Crafted zero-deadspace award-winning systems portfolio featuring hardware-accelerated coordinate canvas, native Web Audio frequency synthesizer, and Raycast-style global command deck (Ctrl+K).

CORE TECHNICAL COMPETENCIES

Browser Internals	Chromium Manifest V3, Content Security Policy (CSP), Main/Isolated World bridging, DOM tree-walking, FOUC mitigation.
Systems & Security	OS hosts routing, Windows Registry Policies (HKLM), SafeSearch VIP mapping, DNS-over-HTTPS policy management, watchdog daemons.
Languages & Web	JavaScript (ES2024), TypeScript, Python, HTML5 Media APIs, CSS Architecture & Tokens, React, Next.js, Node.js, Express, SQLite.
Algorithms & Math	Weighted heuristic scoring, set-intersection optimization, non-linear physical scaling, SHA-256 tiered caching pipelines.